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GLOBAL MALABSORPTION

- Defective absorption of :
  a. Carbohydrate.
  b. Fat : most sensitive & specific, most energy dense nutrient.
  c. Protein.
- Small intestinal mucosal dysfunction.
- Causes :
  a. Celiac disease.
  b. Whipple's disease.
  c. Tropical sprue.

Celiac disease

00:01:01

Features :

- AKA celiac sprue/gluten sensitive enteropathy/non tropical sprue.
- Iceberg phenomenon :
  a. Only 50% presents with classical features. } missed diagnosis } Celiac 2023
  b. D/t physician unawareness.
- HLA DQ2/DQ8 : most important predisposition.
- Proximal small intestinal disease (Duodenum, jejunum).

TYPES

Classical celiac disease :

50% cases presents with symptoms of malabsorption.

Atypical celiac disease :

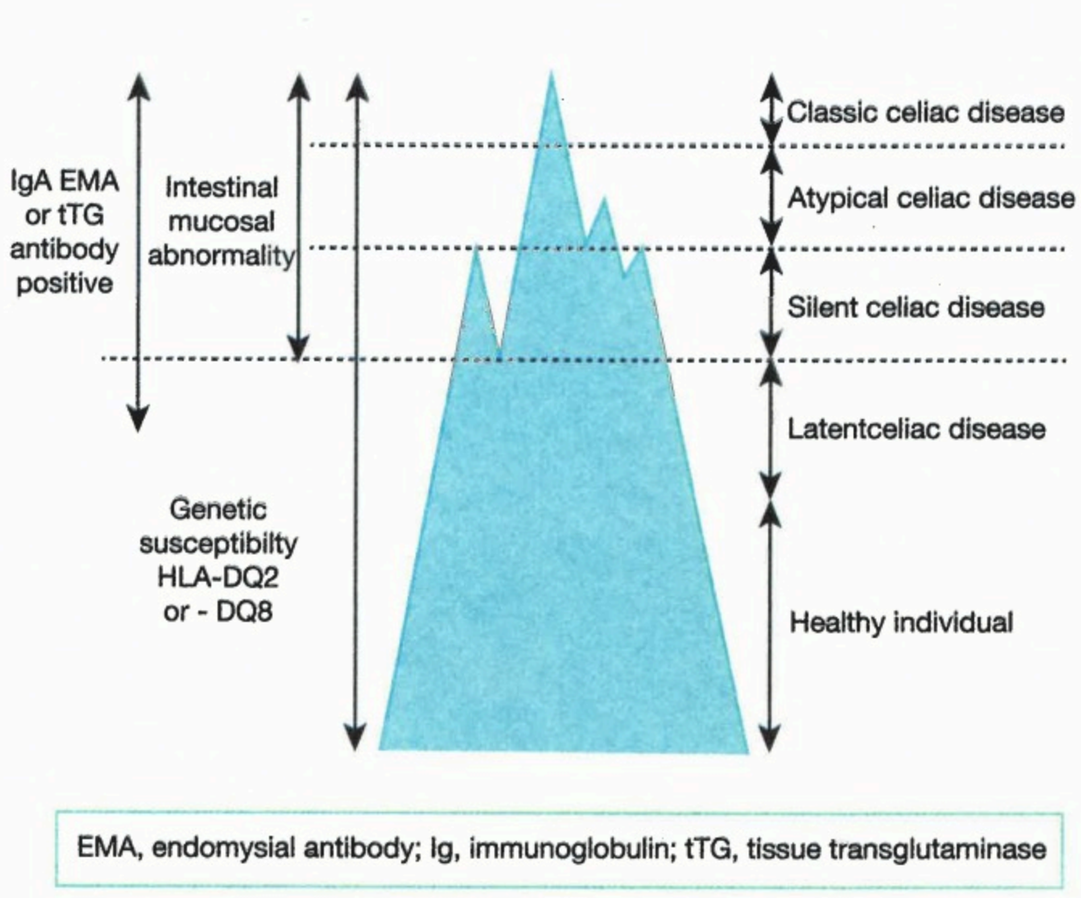
Presents with symptoms other than malabsorption.

Silent celiac disease :

- Asymptomatic.
- Serology positive.

Latent celiac disease :

- Asymptomatic.
- Antibodies negative.
- Presence of HLA susceptibility.



Celiac iceberg and spectrum of celiac disease

PATHOPHYSIOLOGY

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HLA DQ2 or HLA DQ8 Susceptibility
↓
Consumption of gluten containing foods
(wheat, barley, rye, oats)
↓ Proteins
Prolamine (most toxic) — Glutamine
↓
Gliadin + Polymeric IgA
(Chief antigen of gluten) (Produced from MALT)
↓
Forms dimer
↓ Tissue transglutaminase
Deamidation in the intestinal epithelium
↓
Deamidated glutamine dipeptide
↓
Activates T cells
↓
Celiac disease

Note :

- monomeric IgA is produced in bone marrow.
- Vit B_{12}, Mg, bile acid are reabsorbed from distal small intestine → Features not seen in celiac disease

DISEASE PROGRESSION

Gluten-free diet
- 90% → Reverts to normalcy in :
  - Clinically : 3 weeks.
  - Histologically : 6 months.
- 10% → No improvement → Refractory celiac disease → Rx : Steroid

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CLINICAL FEATURES

Classical/typical celiac	Atypical celiac
• Osmotic diarrhea. • Abdominal distension. • Foul smelling stool. • Weight loss. • Symptoms of IBS (30%).	• Short stature, failure to thrive in children. • Fracture, osteopenia : ↓vit D, Ca^{2+} absorption. • Ataxia peripheral neuropathy : ↓vit E. • Iron deficiency anemia (If not due to blood loss). Other features : • Secondary hyperparathyroidism : ↓vit D, Ca. • Amenorrhea, infertility, impotence. • Calcifications in brain seen with secondary hyperparathyroidism. • Hyposplenism : Presence of Howell-Jolly bodies. • Follicular hyperkeratosis : ↓vit A. • Edema & ascites : Protein malabsorption. • Cryptogenic cirrhosis : Raised LFT.

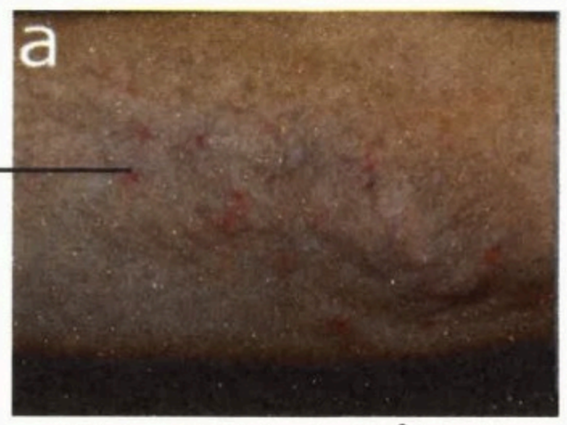
ASSOCIATIONS

Definite associations :

- Dermatitis herpetiformis (most important).

  Features :

  - Papulovesicular lesions in extensor aspect.
  - Granular IgA deposits in dermoepidermal junction.



Dermatitis herpetiformis

- Down syndrome.
- Type I diabetes mellitus.
- IgA nephropathy.
- IgA deficiency.
- Hypersensitivity pneumonitis/bird-fancier lung.
- Lymphocytic colitis.
- Cerebral calcifications with epilepsy.

Probable associations :

- Cryptogenic transaminitis/cirrhosis.
- Idiopathic short stature.
- Refractory iron deficiency anemia.
- Cerebellar ataxia.

DIAGNOSIS

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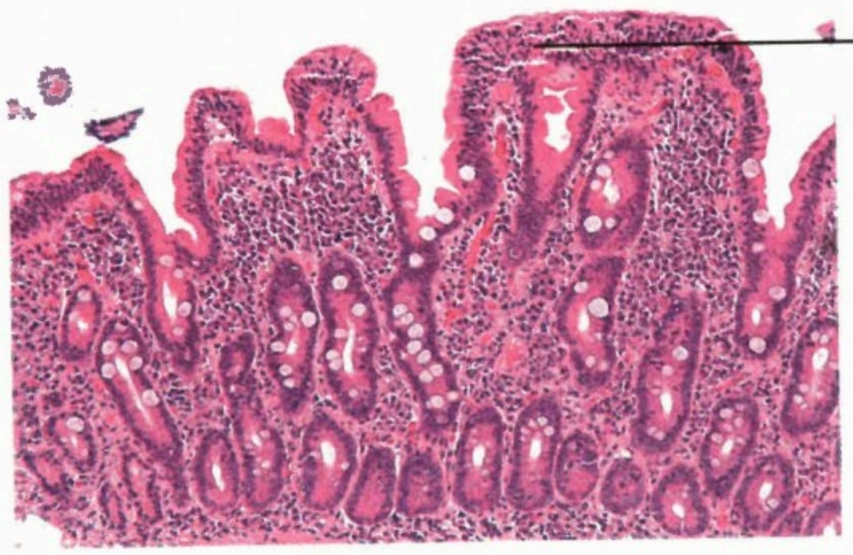
Serology :

	Type	Features
1. Anti-TTG antibody	IgA	• most important for diagnosis, monitoring. • High sensitivity (92%). • High specificity (98%).
2. Anti-Endomysial antibody	IgA	• Done by indirect immunofluorescence. • High specificity (99%). • Poor sensitivity (73%). • Disadvantage : - Subjective interpretation. - Cumbersome procedure.
3. Anti-deamidated glutamine dipeptide	IgG	Done in patients with IgA deficiency.
4. Anti-gliadin	-	• Low sensitivity, specificity. • Obsolete.

Biopsy :



Celiac disease during reversibility

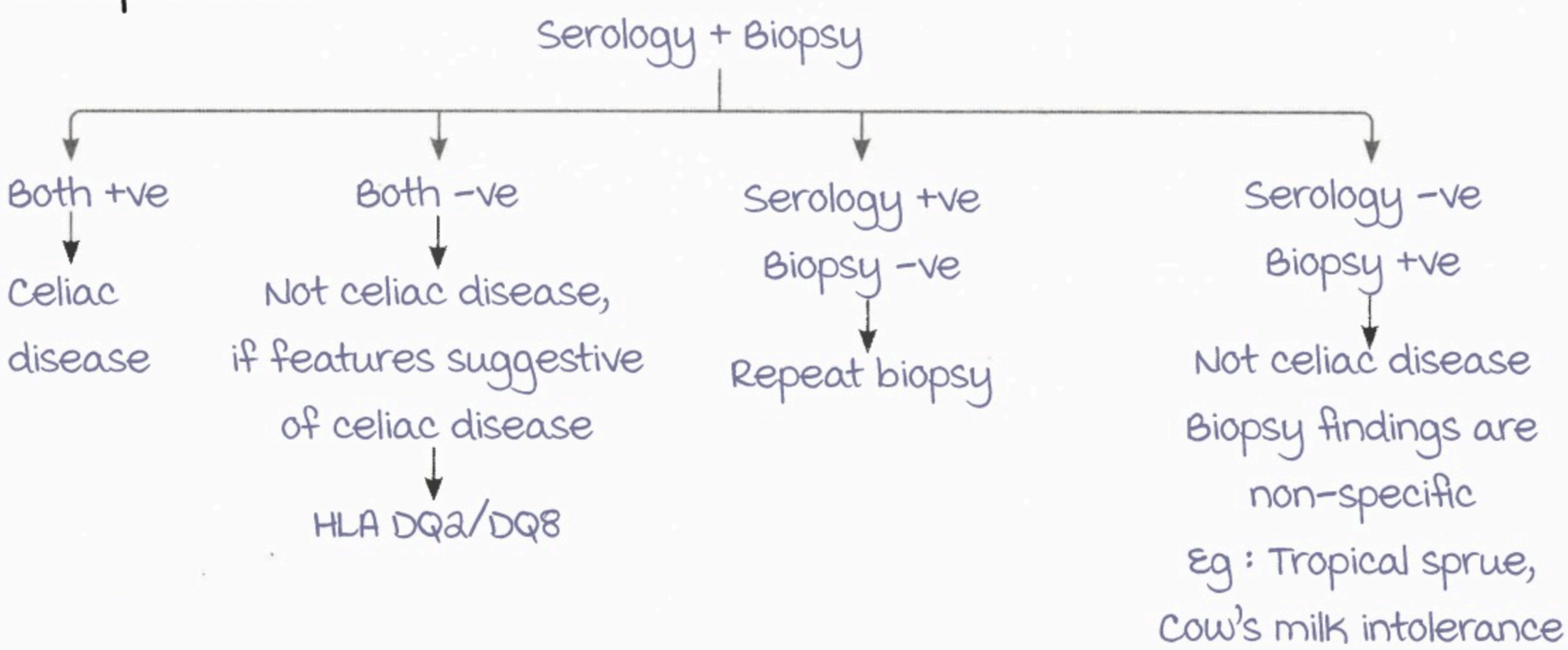


→ Findings localised to mucosa :

1. Total villous atrophy.
2. Crypt hyperplasia.
3. Normal mucosal thickness. } marsh score
4. Lymphocyte infiltrates in lamina propria. } marsh score
5. Vacuolar degeneration of surface epithelium (Non-specific).

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Interpretation :



COMPLICATIONS

Seen in refractory celiac disease.

It is a premalignant condition :

- Enteropathy associated T-cell lymphoma (m/c).
- Small intestinal adenocarcinoma.
- Esophageal squamous cell carcinoma.

TREATMENT

- Diet :

  Rice, corn, maize, potato, soybean based.

  Restriction of wheat, barley, rye, oats.
- Steroids : Refractory celiac disease.

Whipple's Disease

00:25:05

PRESENTATION

- Chronic multisystem involvement (Joints, GIT, CNS, CVS).
- M > F, 50 – 60 yrs.
- Pan-intestinal : Proximal > Distal.
- Tropheryma whipplei : Gram +ve actinobacterium (Cannot be cultured d/t slow doubling time).
- Immune Susceptibility : HLA DRB113, DQB106.
- Dx : Biopsy.

Note :

Conditions diagnosed with biopsy alone :

1. Whipple's disease.
2. Abetalipoproteinemia.
3. Agammaglobulinemia.

CLINICAL FEATURES

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Joints	GIT
• First symptom in >80% cases. • Predominance : Large joints (knee). • Pattern : migratory arthralgia > arthritis.	2-3 yrs after joint features • Diarrhoea (most consistent). • Weight loss. • Abdominal pain + mesenteric lymphadenopathy.
Neuro Whipples (40%)	**CVS**
• m/c neurological manifestation : Rapidly progressive dementia. • m/c specific manifestation : Oculomasticatory myorhythmia > Oculofacial myorhythmia (Associated with supranuclear ophthalmoplegia) : - Oculomasticatory myorhythmia : Pendular nystagmus + masticatory arrhythmias. • Relapse associated with CNS manifestations : Bad prognosis.	• Culture negative endocarditis : Mitral valve > myocarditis > Pericarditis. • Culture negative endocarditis : HACEK group - Haemophilus. - Aggregatibacter actinomycetemcomitans. - Cardiobacter. - Eikenella. - Kingella.

Joint symptoms

Whipple's Tetrad

Weight Loss

Diarrhoea

Abdominal pain + mesenteric lymphadenopathy

Differential diagnosis :

- Joints symptoms : Spondyloarthritis.
- GIT symptoms : TB.

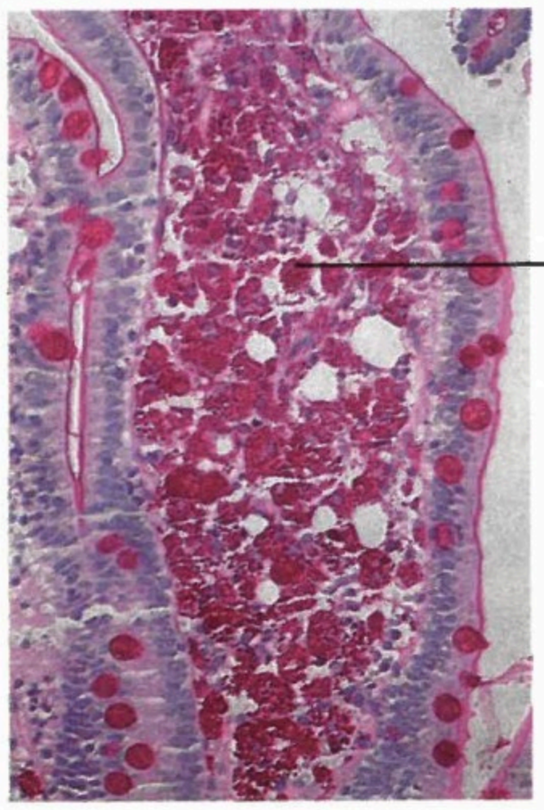
DIAGNOSIS

Upper GI endoscopy.

- Biopsy : IOC.
- PCR : Identification of Tropheryma whipplei.

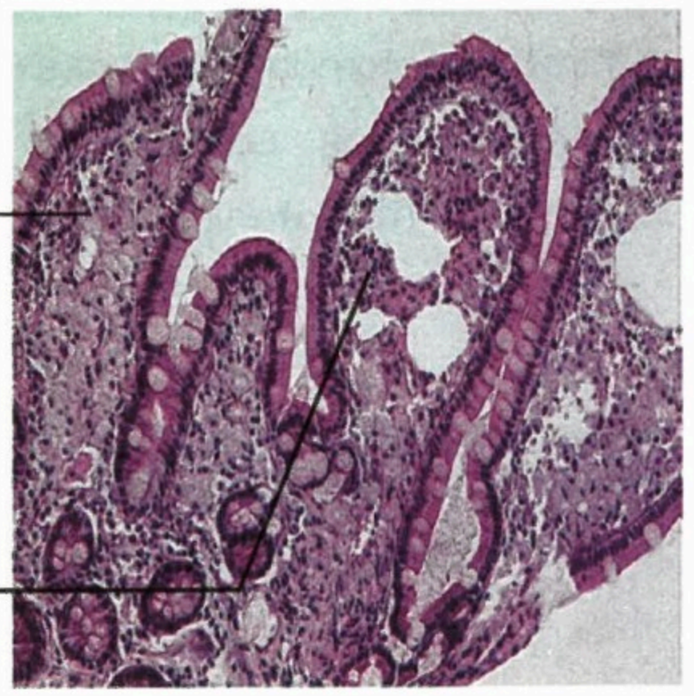
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TREATMENT



- PAS +ve foamy macrophages seen in epithelium, lamina propria, submucosa.
- macrophages contain organism

Enlarged villi



Rx

Induction

I.V Ceftriaxone 2g x 14 days : Best regimen.

- meropenem.
- Penicillin + Streptomycin.

Maintenance

Co-trimoxazole : DOC

- Prevent relapse.
- Good CNS penetration.

Tropical Sprue

00:35:35

m/c cause of malabsorption in India, Bangladesh, Pakistan.

Histological features similar to celiac disease.

PATHOGENESIS

Infective diarrhea

↓

Presence of toxigenic coliform in intestine

(E. coli, Klebsiella)

↓

Fermentation products

(major product : Ethanol)

↓

Structural damage to small intestine

↓

Impairs absorption of folic acid

↓ 2 – 3 weeks

Pan-intestinal involvement

↓

Defective B_{12}, bile acid, mg^{2+} absorption

↓

Clinical features

CLINICAL FEATURES

----- Active space -----

- Chronic small bowel diarrhea (watery, voluminous).
- Steatorrhea.
- Weight loss.
- Folic acid deficiency (most prominent).
- Hyperpigmentation : D/t B_{12} deficiency.


Hyperpigmentation

INVESTIGATIONS

- Biopsy : Features of celiac disease.
- Serology : Negative.
- Stool examination : For ova, cyst, trophozoite (To rule out giardiasis).

TREATMENT

- Folic acid : 5mg daily.
  +
- Tetracycline : 250mg QID.

} 3-6 months

Note :

- Relapse in 20% of cases.
- In Small intestinal bacterial overgrowth (SIBO) :

  Organism → Uses B_{12}
  Organism → Synthesizes folic acid

Bile Acid Diarrhea & Fatty Acid Diarrhea

00:42:18

BILE ACID DIARRHOEA

- AKA cholorrhea.
- Seen in tropical sprue.
- Due to defective FGF-19.

Pathophysiology :

Physiology : Ileal enterocytes → Produce FGF-19 → Sensor to liver → Produce appropriate bile acid.

Pathology : Defective FGF-19 → ↓ bile acid released in the ileum.

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Limited ileal disease :

Any ileal disease
↓
FGF-19 +ve in limited quantity
↓
Liver produces excess bile acid (compensatory)
↓
Bile acid removes Na^+, Cl^+, K^+ from intestinal epithelium
↓
Bile acid diarrhea
No steatorrhea/malabsorption.

FATTY ACID DIARRHOEA

Severe ileal disease
↓
Reabsorption of bile acid
↓
No compensatory release
↓
malabsorption
↓
Fatty acid in colon
↓
Steatorrhea (> 20g)

	Bile acid diarrhea	Fatty acid diarrhea
Ileal bile acid absorption	↓	↓
Fecal bile acid excretion	↑	↑
Bile acid pool size	Normal	↓
Intraduodenal (bile acid)	Normal	↓
Response to cholestyramine	Yes	No
Response to low-fat diet	No	Yes

Short Bowel Syndrome 00:47:20

Length of small intestine (SI) :

- Short bowel syndrome : < 200 cm.
- Very severe short bowel syndrome : < 100 cm.

ETIOLOGY (m/c)

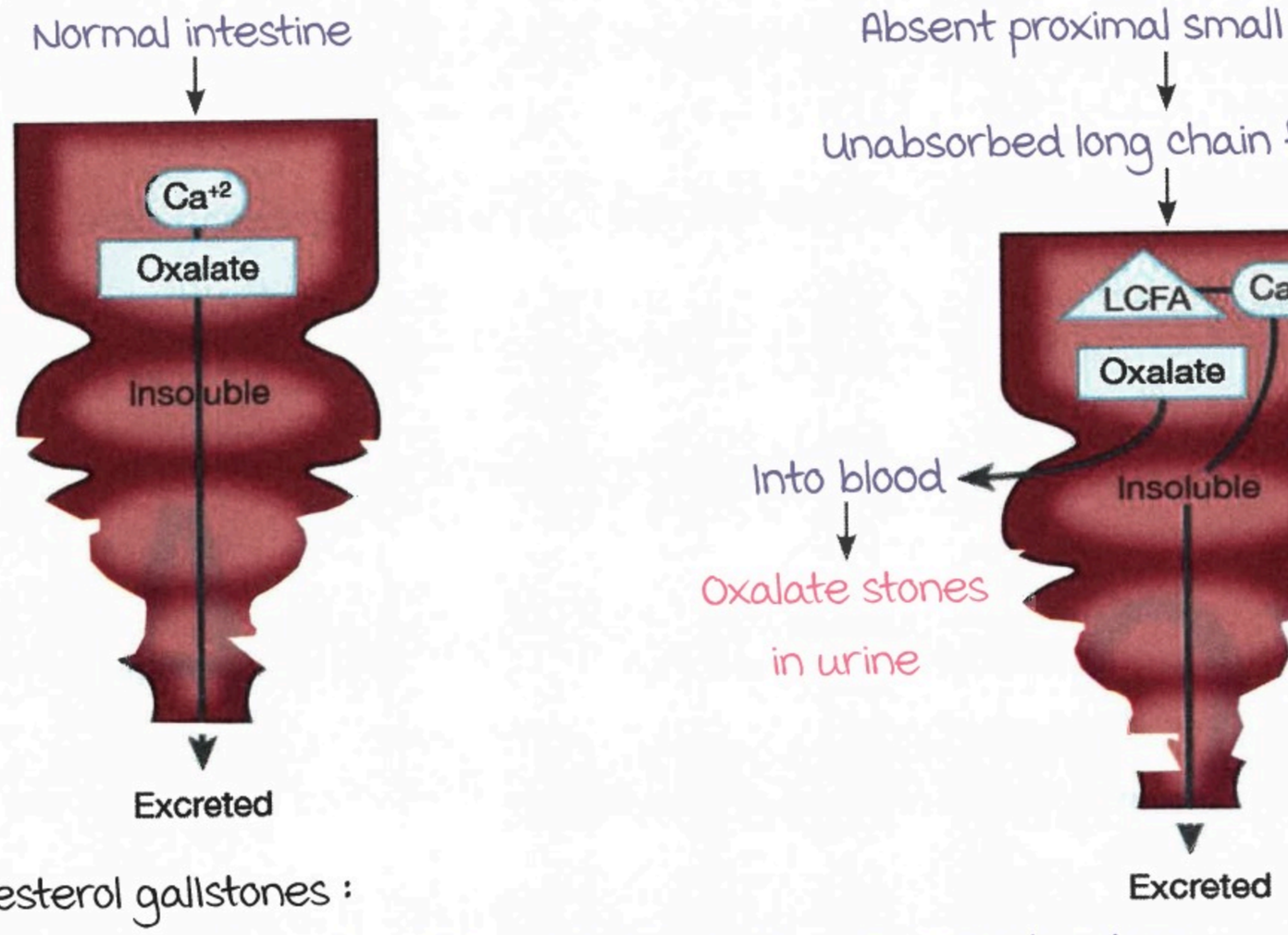
Adult : Crohn's disease

Child : Necrotising enterocolitis.

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CLINICAL PRESENTATION

1. Hyperoxaluria :



2. Cholesterol gallstones :

Absent ileum → Bile acids, vit B_{12} not absorbed → Supersaturation.

3. D-lactic acidosis :

Physiological :

- L-lactate formed.
- Flora of colon : Bacteroides fragilis.

Pathology : Impaired digestion of carbohydrate in small bowel

↓ In colon

Excessive fermentation by gram positive anaerobes

↓ pH in colonic lumen

Bacterial overgrowth (Lactobacillus, Bifidobacterium)

D-lactate (Not metabolised)

↓ Absorbed into blood

D-lactic acidosis → CNS manifestations

4. Hypergastrinemia : Absent small bowel → ↓intestinal hormones → Gastrin uninhibited.

5. Hypomagnesemia : No Mg^{2+} absorption from ileum.

6. ↑Gastric emptying : Defective ileal brake.

Physiology : ↑unabsorbed fat enters ileum $\xrightarrow{\text{GLP-1, GLP-2, peptide YY}}$ ↓motility → ↓gastric emptying (ileal break).

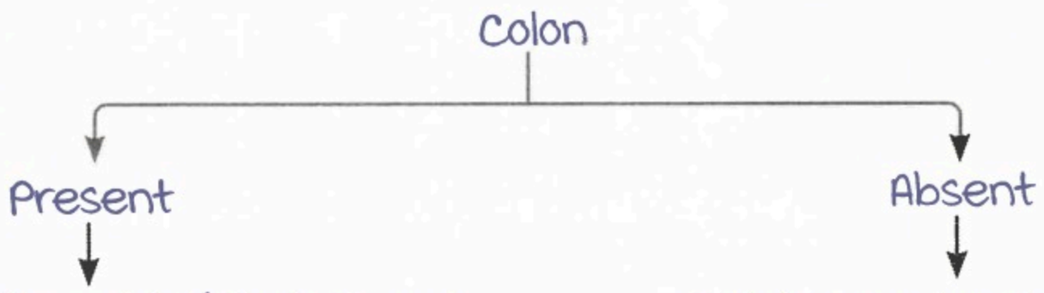
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7. Deficiencies of :
   - Fe : In absent duodenum > Jejunum.
   - Ca^{2+}, folate, vit A, D, E : In absent jejunum.
   - Vit K : Rare d/t synthesis (Absent ileocecal valve → migration of bacteria).

8. SIBO (Small Intestinal Bacterial Overgrowth) : D/t loss of ileocecal valve.

MANAGEMENT

Diet :



- medium chain fatty acids (Lipolysis not required for its absorption).
- Complex carbohydrates (Absorbed in colon as short chain fatty acids).
- Protein : 1 g/kg/day.

Medical : Tedaglutide (Oral GLP-2 analogue).

Small Intestinal Bacterial Overgrowth syndrome (SIBO) 00:55:20

AKA blind loop syndrome, stagnant bowel Syndrome.

PHYSIOLOGY

Proximal small intestine : No bacteria.

Ileum : Enterobacteriaceae.

Colon : Anaerobes.

PATHOLOGY

Bacterial overgrowth :

- Non toxigenic anaerobes (Bacteroides fragilis, Enterobacteriaceae, Pseudomonas).
- In the proximal small intestine.

ETIOLOGY

1. Anatomical defect : Fistula, Ileocecal valve removal.
2. Functional and motility disorders :
   - Irritable bowel syndrome (IBS) : Defective migratory motor complex.
   - Scleroderma.
   - Diabetes mellitus.
   - Amyloidosis.

3. Hypochlorhydria : In ↑intake of PPI (proton pump inhibitors).
4. Immunodeficiency :
   - Common variable immune deficiency (CVID).
   - IgA deficiency.
5. Multifactorial : HIV.

----- Active space -----

CLINICAL PRESENTATION

Bacteria in lumen → Compete for host nutrients → Consume B_{12}, proteins, vitamins.

1. Deficiency of Vit B12.
2. Macrocytic anemia.
3. Steatorrhea.
4. Unexplained diarrhoea.
5. Bloating, belching :
   - m/c symptom.
   - D/t unabsorbed carbohydrates.
6. Occult malabsorption : Some cases present with only weight loss.

	SIBO	Tropical sprue
Organisms in SI	Non toxigenic anaerobes	Toxigenic aerobes
Vit B12 deficiency	⊕ (Prominent)	⊕
Folate deficiency	⊖	⊕ (Prominent)
Vit K deficiency	⊖	⊕

INVESTIGATIONS

1. Jejunal fluid aspirate : >10^3 CFU/mL (Gold standard).
2. Lactulose hydrogen breath test :
   - < 90 mins (D/t bacteria in SI).
   - Easy to perform (Preferred).
3. Urinary indican levels : Obsolete.
4. Serum Vit B_{12} : Decreased.
5. S. folate and Vit K : Increased.

MANAGEMENT

Antibiotics : 7–10 days.

- Tab rifaximin 550 mg BD (DOC).
- Tab rifaximin 550 mg + Tab neomycin 500 mg BD.

On persistence/recurrence :

- Amoxicillin-clavulanic acid 500 mg TID.
- Tetracycline 250 mg QID.
- metronidazole 250 mg TID.

Avoid macrolides.